PARUL INSTITUTE OF ENGINEERING & TECHNOLOGY
Department of AI & Data Sciences | Subject: Linear Algebra
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Assignment 3: Matrix Spectral Decomposition
Question 1: Find all eigenvalues of the 2x2 matrix A = [[4, 1], [2, 3]].

Solution Steps (Handwritten Derivation):

1. Write the characteristic equation:
   det(A - lambda * I) = 0
   | 4 - lambda 1 |
   | 2 3 - lambda | = 0

2. Compute the 2x2 determinant:
   (4 - lambda)(3 - lambda) - (1 * 2) = 0
   12 - 4*lambda - 3*lambda + lambda^2 - 2 = 0
   lambda^2 - 7*lambda + 10 = 0 <-- Characteristic Polynomial

3. Factorize the quadratic polynomial:
   (lambda - 5)(lambda - 2) = 0

4. Solve for roots (Eigenvalues):
   lambda_1 = 5
   lambda_2 = 2

Conclusion: The matrix has two distinct real eigenvalues: lambda in {5, 2}.